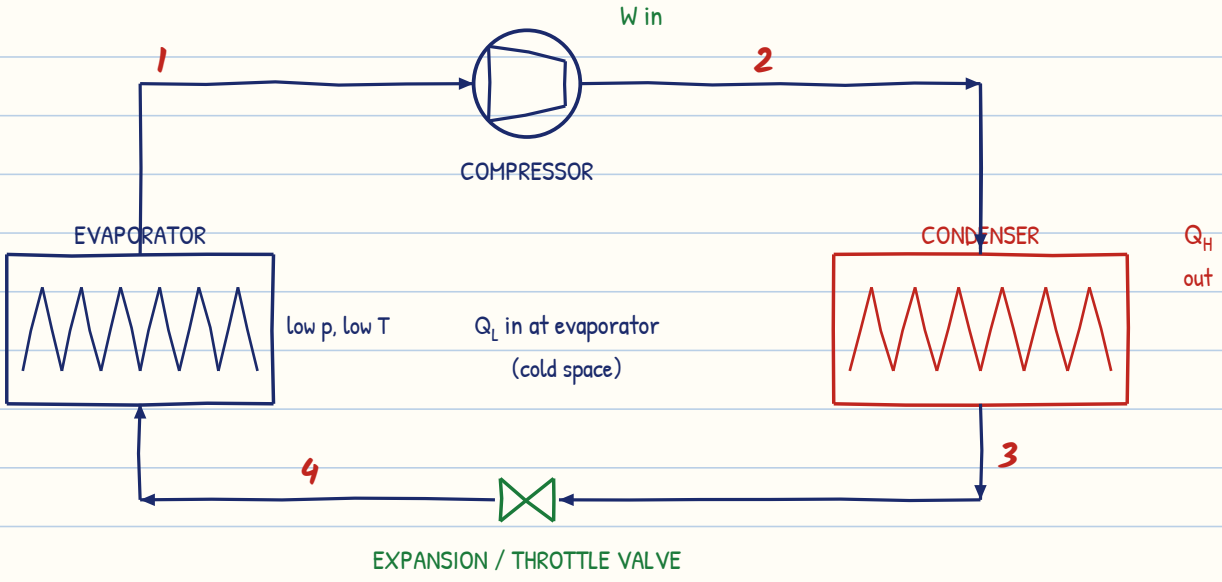
Syllabus: VCR system – basic cycle, working principle and components, COP, T-s and p-h charts, expander vs throttling, effect of sub-cooling and super-heating, cycle analysis, actual cycle, influence of parameters, p-h chart, numericals. Refrigerants – properties, classification, nomenclature, secondary refrigerants, lubricants, ozone depletion, global warming, newer refrigerants.

I. Vapour Compression Refrigeration (VCR) System

- Most widely used refrigeration system (domestic fridge, AC, cold storage, chillers).
- Working fluid is a **refrigerant** that changes phase: it absorbs the latent heat while **evaporating** at low temperature and rejects heat while **condensing** at high temperature.
- Mechanical energy (compressor) is the input; used for capacities from 50 W to many MW.

Essential components

- **Compressor** – raises pressure and temperature of vapour (reciprocating, rotary, scroll, screw, centrifugal). It circulates the refrigerant (the "heart" of the plant).
- **Condenser** – desuperheats, condenses and (sub-cools) the vapour; rejects Q_H to air or water.
- **Expansion device** – throttle valve / capillary tube / thermostatic expansion valve (TEV); reduces pressure from p_c to p_e and controls refrigerant flow.
- **Evaporator** – refrigerant boils at low pressure, absorbs Q_L from the cold space.
- Accessories: receiver, filter-drier, sight glass, accumulator, oil separator, pressure cut-outs.



Components in detail

Component	Types	Notes
Compressor	Reciprocating, rotary, scroll, screw, centrifugal	Recip. up to ~100 TR; centrifugal for > 200 TR chillers
Condenser	Air-cooled, water-cooled, evaporative	Air-cooled: small units; water: cooling tower; evaporative: ammonia plants
Evaporator	Dry-expansion, flooded, shell & tube, bare tube / finned coil	Flooded gives higher h, dry-expansion uses TEV
Expansion device	Capillary tube, TEV, automatic EV, float valve, electronic EV	Capillary: fixed orifice, domestic; TEV keeps constant superheat

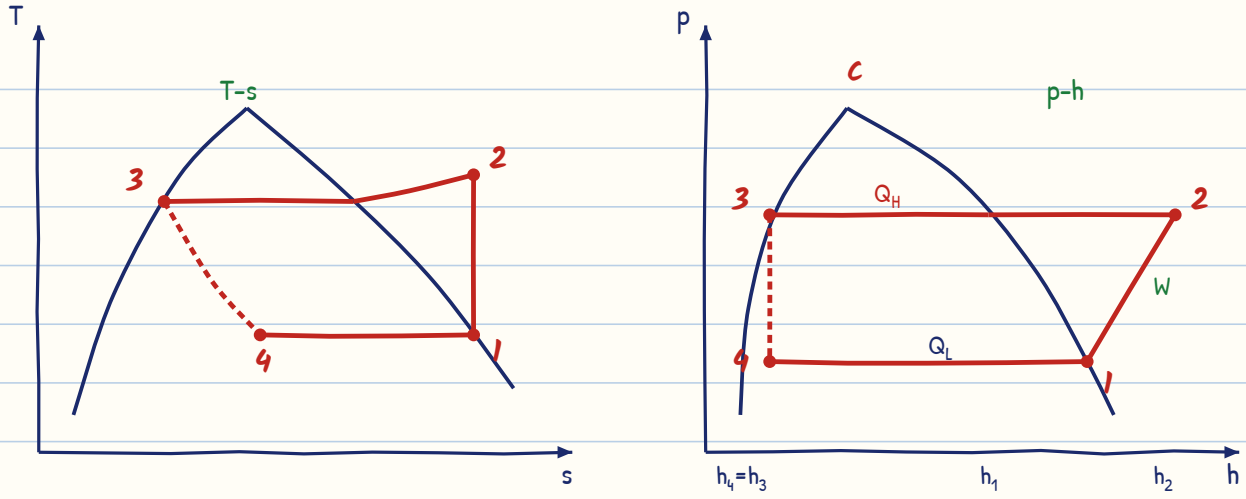
- **Receiver** stores liquid so that load variations are handled; **filter-drier** removes moisture and dirt; **accumulator** prevents liquid reaching compressor; **oil separator** returns oil to crankcase.
- **Safety controls:** HP/LP cut-out, oil pressure switch, thermostat, relief valve.

Hermetic, semi-hermetic and open compressors

- **Hermetic:** motor + compressor sealed in one shell (fridges, ACs) – no shaft seal leak, noisy-free, but not serviceable.
- **Semi-hermetic:** bolted shell, can be opened for service. **Open type:** motor outside, shaft seal, used for NH_3 (motor cannot be exposed to NH_3).

Working principle (simple saturated cycle)

- **1-2** Isentropic compression of dry saturated vapour from p_e to p_c. Vapour becomes superheated: T rises, work W input.
- **2-3** Constant pressure cooling and condensation in condenser (desuperheating 2-2', condensation 2'-3). Heat Q_H rejected. State 3 is saturated liquid.
- **3-4** Throttling (isenthalpic expansion) through the valve: p falls to p_e, temperature falls to T_e, part of the liquid flashes to vapour. $h_3 = h_4$.
- **4-1** Constant pressure, constant temperature evaporation. Refrigerant absorbs Q_L (latent heat) from the cold space.



Cycle analysis (per kg of refrigerant)

Refrigerating effect $q_L = h_1 - h_4 = h_1 - h_3$

Compressor work $w = h_2 - h_1$

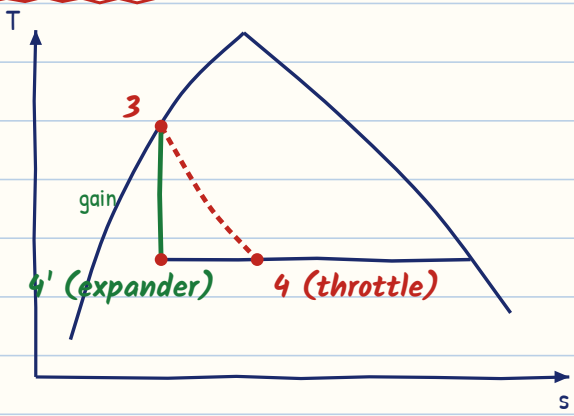
Heat rejected $q_H = h_2 - h_3 = q_L + w$

$$COP = (h_1 - h_4) / (h_2 - h_1) = (h_1 - h_3) / (h_2 - h_1)$$

- Mass flow rate $m = Q_L / (h_1 - h_4) = (TR \times 3.5167) / (h_1 - h_4)$ kg/s
- Compressor power $P = m (h_2 - h_1)$; Condenser load = $m (h_2 - h_3)$.
- Dryness fraction at 4: $x_4 = (h_4 - h_{f4}) / (h_{fg4})$ – the **flash gas** fraction, which gives no cooling.
- Volume flow at compressor inlet $V = m v_1$; swept volume = V/η_{vol}.

Note: Compare with the Carnot cycle: the VCR cycle replaces the wet compression with dry compression and the expander with a throttle – practical but a little less efficient.

2. Expander vs Throttling



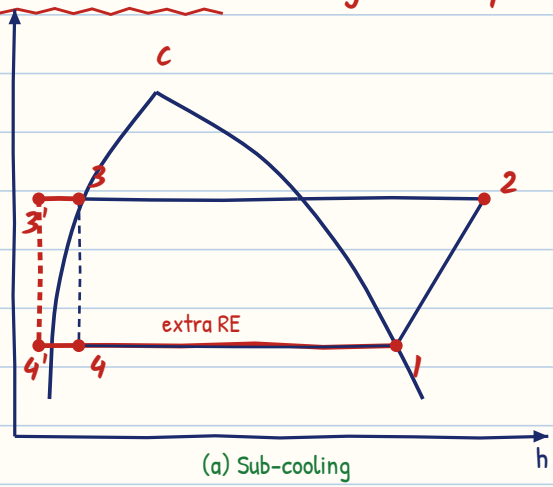
Throttle (3–4): h = const, irreversible, no work recovered, simple, cheap.

Expander (3–4'): s = const, work recovered, RE rises by ($h_4 - h_{4'}$), but complex & costly; liquid wet-expansion problems.

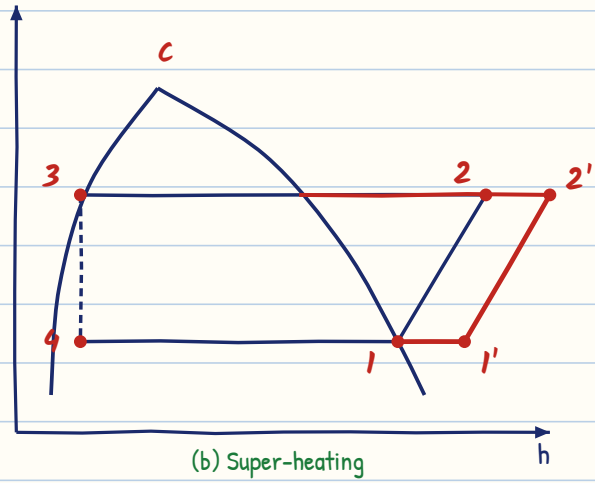
Gain is small (≈ 2–5 %) → throttle is used.

- With an expander the work recovered is ($h_3 - h_{4'}$) and COP improves, but liquid expansion makes work very small because the liquid is nearly incompressible; friction and cost cancel the gain.
- Throttle valve / capillary has no moving parts, is cheap and controls flow – therefore used in all practical VCR plants. (Expanders are used in CO_2 transcritical and some cryogenic plants where the gain is large.)

3. Effect of Sub-cooling and Super-heating



(a) Sub-cooling



(b) Super-heating

Sub-cooling (3 → 3')

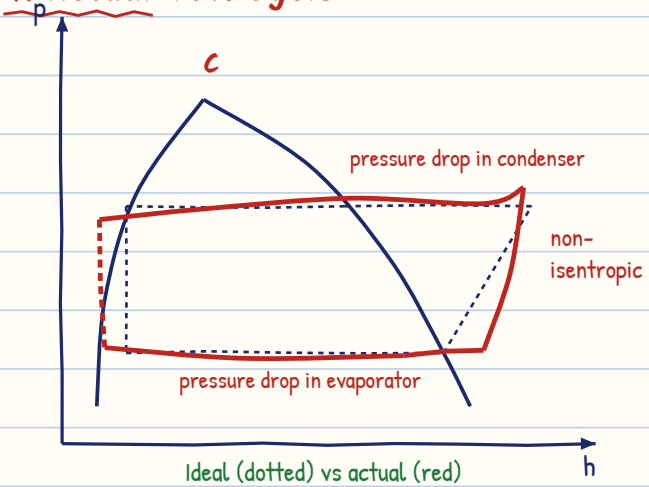
- Liquid cooled below saturation temperature at condenser pressure (by sub-cooler, liquid-suction heat exchanger, or extra length in condenser).
- Refrigerating effect increases ($h_1 - h_{4'}$), compressor work unchanged → **COP increases**. Also prevents flash gas before the valve.

Super-heating (1 → 1')

- Vapour leaves the evaporator slightly above saturation (5-10 K) – this protects the compressor from **liquid slugging** and helps TEV control.
- **Useful super-heating** (inside the cooled space) increases RE and COP slightly; **useless super-heating** (in suction line outside) increases work with little gain.
- Excessive super-heating raises discharge temperature and may break down oil.

Change	RE	Work	COP
Sub-cooling	Increases	Same	Increases
Super-heating (useful)	Increases	Increases	Increases / almost same
Super-heating (useless)	Same	Increases	Decreases
Higher evaporator temperature	Increases	Decreases	Increases
Higher condenser temperature	Decreases	Increases	Decreases

4. Actual VCR Cycle



Actual cycle differs from ideal:
- suction superheat (1-1')
- pressure drops in lines, valves
- compression not isentropic
- sub-cooling in condenser
- heat gain in suction line
- valve losses, leakage

- Real compression has friction and heat transfer → entropy increases, $\eta_{isen} = (h_{2s} - h_1)/(h_2 - h_1) \approx 0.7\text{-}0.85$.
- **Volumetric efficiency** $\eta_{vol} < 1$ due to clearance, valve leakage, wire drawing and heat transfer.
- Superheating at inlet, sub-cooling at outlet of condenser, pressure drop everywhere: the evaporator must be at a lower p and the condenser at a higher p than ideal cycle → more work.

- Real COP is only 50-65 % of the ideal cycle value.

Combined sub-cooling + super-heating cycle

Let h_1 = enthalpy after super-heating, h_2 = after compression from 1, h_3 = enthalpy of sub-cooled liquid = h_4.

$$COP = (h_1 - h_3) / (h_2 - h_1) \text{ ; isentropic efficiency } \eta = (h_{2s} - h_1) / (h_2 - h_1)$$

- Sub-cooled liquid enthalpy can be approximated by h_f at its own temperature: $h = h_f(T_{sub})$.
- Super-heated vapour enthalpy: $h_1 \approx h_g + c_{pv} (T_{sup} - T_{sat})$.

Liquid-suction heat exchanger (LSHX)

- A small counterflow exchanger that sub-cools liquid from condenser using the cold suction vapour: heat balance $h_3 - h_{3'} = h_{1'} - h_1$.
- Benefits: avoids flash gas, ensures dry suction; effective with R134a, R12; poor for NH_3 (high discharge temp).

5. Influence of Various Parameters on System Performance

Evaporator pressure / temperature (T_e)

- Raising T_e: RE ↑, specific volume at suction ↓ (so refrigerant flow rises), compressor work per kg ↑ only slightly → **COP ↑ (about 2-4 % per K)**.
- Lowering T_e reduces capacity sharply, needs a bigger compressor, and may lead to frosting.

Condenser pressure / temperature (T_c)

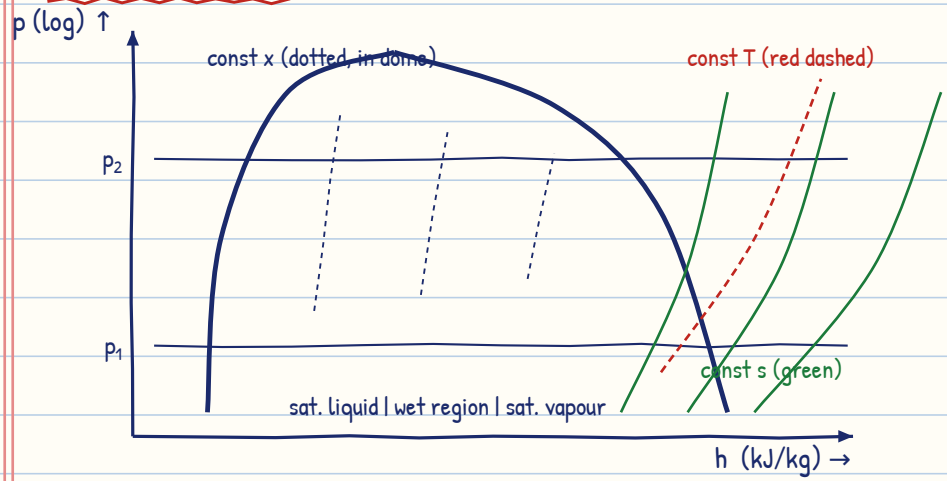
- Raising T_c: work ↑, RE ↓ (due to more flash gas), discharge temp ↑ → **COP ↓, capacity ↓**.
- Keep condenser clean; use cooling water or large air flow; avoid non-condensable gases.

Others

- **Sub-cooling and super-heating** as above. **Compressor speed / clearance:** affects η_{vol}.
- **Type of refrigerant:** latent heat, specific volume decide the flow rate and size.
- **Moisture and oil in system:** choke the valve, reduce heat transfer.

> **Note:** For every 1 K rise in condensing temperature the power input increases by roughly 2-3 %.

6. Construction and Use of p-h Chart



Lines on p-h chart:

- Saturation dome (x = 0, x = 1)
- Isobars – horizontal
- Isenthalpic – vertical
- Isotherms – vertical in liquid, horizontal in dome, steep in vapour
- Isentropes – rising to right
- Constant volume & quality lines

- The p-h (Mollier) chart plots **pressure (log scale, y)** against **enthalpy (x)**, each refrigerant has its own chart.
- It gives every enthalpy needed (h_1, h_2, h_3, h_4) directly as **horizontal lengths** – hence the cycle is easy to calculate.
- **Construction (steps):** (i) draw the saturated liquid and vapour curves meeting at the critical point; (ii) draw isobars (horizontal) at various pressures; (iii) join equal enthalpy points to draw vertical isenthalpic lines; (iv) draw isotherms: vertical in sub-cooled liquid, horizontal in wet region, falling steeply in superheated zone; (v) add isentropic lines (s = const), constant volume and dryness fraction lines.

Using the chart for a simple saturated cycle

- Locate evaporator pressure line; mark 1 on the saturated vapour line.
- From 1 follow the isentrope up to p_c to get 2; read h_2.
- From 2 go horizontally left to the saturated liquid line to get 3; drop vertically to p_e to get 4 ($h_4 = h_3$).
- Read h_1, h_2, h_3 and use the COP formula.

7. Numerical Problems

Problem 1

A refrigerator using R-134a works between −10 °C (evaporator) and 40 °C (condenser) on a standard saturated cycle. From tables: h_1 = 392.3, h_2 = 424.0, $h_3 = h_4$ = 256.4 kJ/kg (values approximate). Find RE, work, COP, and the mass flow & power for 5 TR.

RE = 392.3 - 256.4 = **135.9 kJ/kg** ; w = 424.0 - 392.3 = **31.7 kJ/kg**

COP = 135.9/31.7 = **4.29**

m = 5 × 3.5167 / 135.9 = **0.129 kg/s** ; P = 0.129 × 31.7 = **4.10 kW**

Condenser heat = m (h_2 - h_3) = 0.129 × 167.6 = 21.7 kW (check: 17.58 + 4.10)

Problem 2 (effect of sub-cooling)

In problem 1 the liquid is sub-cooled to 35 °C (h = 249.0 kJ/kg). Find new COP. (same compressor work approx.)

RE = 392.3 - 249.0 = 143.3 kJ/kg ; COP = 143.3/31.7 = **4.52** → increase of 5.4 %.

Problem 3 (given enthalpies)

A VCR plant of 10 TR has h_1 = 380, h_2 = 430, h_3 = 240 kJ/kg. Find COP, mass flow, power, condenser load.

RE = 140 ; w = 50 ; COP = 140/50 = **2.8**

m = 35.167/140 = **0.251 kg/s** ; P = 0.251 × 50 = **12.56 kW** ; Q_H = 0.251×190 = **47.7 kW**

Problem 4 (volumetric flow)

In problem 3, suction specific volume is 0.07 m^3/kg, η_{vol} = 0.8. Swept volume = m v_1/η_{vol} = 0.251 × 0.07/0.8 = **0.022 m^3/s** = 79 m^3/h.

Problem 5 (relative COP)

Carnot COP for −10 °C and 40 °C = 263/50 = 5.26. Relative COP for problem 1 = 4.29/5.26 = **0.82**.

Problem 6 (ammonia plant, data given)

A 20 TR ammonia plant has h_1 = 1450, h_2 = 1670, h_3 = h_4 = 340 kJ/kg. Find COP, mass flow, power.

RE = 1450 - 340 = 1110 kJ/kg ; w = 220 kJ/kg ; COP = 1110/220 = **5.05**

$m = 20 \times 3.5167/1110 = \mathbf{0.0634\ kg/s}$; $P = 0.0634 \times 220 = \mathbf{13.9\ kW}$

Problem 7 (super-heating + sub-cooling)

An 8 TR plant: $h_1 = 395$, $h_2 = 432$ (after super-heat), $h_3 = h_4 = 245$ kJ/kg (sub-cooled).

$RE = 150$; $w = 37$; $COP = 150/37 = \mathbf{4.05}$; $m = 28.13/150 = \mathbf{0.1876\ kg/s}$; $P = 0.1876 \times 37 = \mathbf{6.94\ kW}$

Problem 8 (isentropic efficiency of compressor)

In problem 1, the compressor isentropic efficiency is 80 %. Actual $h_2 = 392.3 + (424.0 - 392.3)/0.8 = 431.9$ kJ/kg.

$w_{actual} = 39.6$ kJ/kg ; $COP_{actual} = 135.9/39.6 = \mathbf{3.43}$ (instead of 4.29)

Problem 9 (power per TR)

For problem 1: power per TR = $4.10/5 = \mathbf{0.82\ kW/TR}$, EER = $3.412 \times 4.29 = 14.6$.

8. Refrigerants

A refrigerant is the working fluid that absorbs heat by evaporating and rejects heat by condensing.

Desirable properties

- **Thermodynamic:** high latent heat; low boiling point at atmospheric pressure; evaporator pressure above atmosphere (no air leak-in); moderate condenser pressure; low critical-point issues (critical T well above T_c); low freezing point; low specific volume of vapour; high COP; low discharge temperature.
- **Physical / chemical:** chemically stable, non-corrosive, non-reactive with oil and metals; good heat transfer (high conductivity), low viscosity; miscible with oil as required.
- **Safety:** non-toxic, non-flammable, non-explosive; easy leak detection; low cost and easily available.
- **Environment:** zero ODP and low GWP.

Classification

Group	Examples	Remarks
Halocarbons – CFC	R11, R12, R113	ODP high, phased out
HCFC	R22, R123, R141b	ODP small, being phased out
HFC	R134a, R32, R125, R152a	ODP 0, GWP moderate
HFO	R1234yf, R1234ze	GWP < 1, mildly flammable
Inorganic	R717 (NH_3), R718 (water), R744 (CO_2), R729 (air)	Natural refrigerants
Hydrocarbons	R290 (propane), R600a (isobutane), R1270	Flammable, zero ODP, tiny GWP
Azeotropes	R500, R502, R507A	Behave as pure fluid (R-5xx)
Zeotropes (blends)	R404A, R407C, R410A	Temp glide (R-4xx)

Nomenclature (ASHRAE numbering)

- For R-**XYZ** (hydrocarbon derivatives): X = C - 1 (omit if zero), Y = H + 1, Z = number of F atoms. Remaining valence is Cl.

R12 = CCl_2F_2 (C-1 = 0, H+1 = 1, F = 2)

R22 = $CHClF_2$ (0, 2, 2) R134a = $C_2H_2F_4$ (1, 3, 4; 'a' = isomer)

R32 = CH_2F_2 (0, 3, 2) R11 = CCl_3F (0, 1, 1)

- Inorganic: R-**7XX** with XX = molecular mass: NH_3 = R717, water = R718, CO_2 = R744.
- Blends: R-4XX (zeotropic), R-5XX (azeotropic). R-4xx with letter: composition, e.g. R410A = R32/R125 (50/50).
- Isobutane = R600a, propane = R290.

NH_3 versus R134a (comparison)

Property	R717 (NH_3)	R134a
Latent heat at −15 °C	~1315 kJ/kg	~210 kJ/kg
Boiling point at 1 atm	−33.3 °C	−26.3 °C
Toxicity / flammability	Toxic, B2 (mild flammable)	Non-toxic, A1
Compatibility	Not with copper	Compatible
Application	Industrial, cold storage	Domestic, car AC
ODP / GWP	0 / 0	0 / 1430
Leak detection	Smell, sulphur candle	Halide torch, electronic

Mixtures

- **Azeotrope:** boils at a constant temperature like a pure fluid (R502 = R22 + R115; R507A). **Zeotrope:** composition changes during phase change, so there is temperature **glide** (R407C glide ≈ 5 K).
- R410A is a near-azeotrope with small glide, runs at ≈ 60 % higher pressure than R22.

Secondary refrigerants (brines)

- Heat-carrying fluid cooled by the primary refrigerant, then circulated to the load (indirect system) – used in large plants, ice rinks, breweries, to reduce primary charge.
- Examples: water (above 0 °C), **NaCl brine** (eutectic 23.1 %, −21.1 °C), **$CaCl_2$ brine** (29.9 %, −55 °C), **ethylene / propylene glycol** (non-toxic, for food), methanol, CO_2.
- Needs: low freezing point, high specific heat, low viscosity, non-corrosive (inhibitors), cheap.

Lubricants

- Compressor oil lubricates, seals, cools. Properties: low pour point, correct viscosity, high flash point, chemical stability, miscibility with refrigerant, low wax formation (floc point).
- **Mineral oil** (CFC, HCFC, NH_3), **alkylbenzene**, **POE** (polyol-ester, for HFC), **PAG** (auto AC with R134a), **PAO**. HFCs are not miscible with mineral oil – use POE.

Ozone depletion and global warming

- Chlorine from CFC/HCFC in the stratosphere destroys ozone: $Cl + O_3 \rightarrow ClO + O_2$. **ODP** is relative to R11 = 1. R12 = 1, R22 = 0.055, R134a = 0.
- **GWP** relative to CO_2 = 1 (100 yr): R12 ≈ 10 900, R22 ≈ 1 810, R134a ≈ 1 430, R410A ≈ 2 090, R32 ≈ 675, R1234yf ≈ 4, R290 ≈ 3, NH_3 = 0.

- **Montreal Protocol (1987)** – phase-out of CFCs and HCFCs. **Kigali Amendment (2016)** – phase-down of HFCs. **Kyoto Protocol** – greenhouse gases.

Refrigerant	ODP	GWP	Notes
R12	1.0	~10900	banned
R22	0.055	~1810	phasing out
R134a	0	~1430	fridge, car AC
R410A	0	~2090	split AC
R32	0	~675	new split AC
R290	0	~3	flammable
R717 (NH_3)	0	0	toxic, industrial

Newer refrigerants

- **R32** (lower GWP, A2L), **R1234yf** (car AC), **R1234ze** (chillers), **R290 / R600a** (domestic), **R744 (CO_2)** transcritical systems, **R454B / R513A** (low-GWP blends replacing R410A / R134a), natural refrigerants NH_3.
- Trend: ZERO ODP, low GWP, mildly flammable (A2L) with safety standards.

Safety classification of refrigerants (ASHRAE 34)

- Letter = toxicity: **A** = lower toxicity, **B** = higher toxicity. Number = flammability: **1** = no flame propagation, **2L** = mildly flammable, **2** = flammable, **3** = highly flammable.

Class	Example
A1	R134a, R410A, R22, CO_2
A2L	R32, R1234yf, R1234ze
A3	R290, R600a
B2L	R717 (ammonia)

Leak detection and charging

- Halide torch (flame turns green), soap bubble test, electronic detector, UV dye; ammonia – sulphur candle (white smoke) or litmus paper.
- Always **evacuate** the system with a vacuum pump (below 500 microns) before charging – removes air and moisture.
- Charge by weight; check superheat (TEV) or sub-cooling (capillary systems).

Example: effect of T_e on COP

For R134a at T_c = 40 °C: COP(T_e = −10 °C) ≈ 4.3, COP(T_e = −20 °C) ≈ 3.2, COP(T_e = 0 °C) ≈ 6.0. So raising evaporator temperature by 10 K increases COP by about 40 %. This is why chillers are run at the highest possible evaporator temperature.

9. Quick Revision and Exam Questions

- COP = $(h_1 - h_4)/(h_2 - h_1)$; m = TR×3.5167/RE; $h_3 = h_4$.
- Sub-cooling ↑ COP; ↑T_e ↑ COP; ↑T_c ↓ COP.
- R-number rule (C-1)(H+1)(F); ODP of R11 = 1; GWP of CO_2 = 1.

Likely questions

- Draw VCR cycle on T-s and p-h charts; derive COP; list components with functions. (long)
- Compare expander and throttle valve. (short)
- Discuss effect of sub-cooling, super-heating, evaporator and condenser temperatures on COP. (long)
- Explain actual VCR cycle and its deviations. (long)
- Properties of a good refrigerant; nomenclature of R12, R22, R134a, R717; secondary refrigerants. (long)
- ODP, GWP, Montreal and Kigali, newer refrigerants. (short)
- Numerical on VCR with given enthalpies: COP, mass flow, power.